

Jimmy Lee 李兆康

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ACADEMIC POSITIONS

Postdoctoral Fellow in Economics

Global Poverty Research Lab, Northwestern University

Sept 2024 – Present

Visiting Scholar

Center for International Development, Harvard University

Jan 2025 – June 2026

RESEARCH FIELDS

Development Economics, Program Evaluation and Scaling

Focus—Synergies between Agriculture, Rural Education, and Digital Technologies

Family Economics, Organizational Economics

Focus—Intergenerational Dynamics in Rural Household and Community Decision Making

Behavioral Economics with applications in program evaluation, agriculture, and education

Focus—Modeling Curiosity Through Contextual Shifts in Mindsets: Endogenous Model Uncertainty and Information Acquisition Under Cognitive Limits

EDUCATION

Ph.D., Economics, Northwestern University

2024

M.Phil., Economics, The Chinese University of Hong Kong

2017

B.SSc., Economics, The Chinese University of Hong Kong

2015

JOB MARKET PAPER

“Miscategorization as an Information Friction? Activating School-Based Agricultural Extension in Liberia with Parental Engagement”

(Funded by AFD Fund for Innovation in Development, USAID-DIV, Wellspring Foundation, and FAO)

Schools in developing countries often have weak ties with parents. This paper studies the extent to which development program success hinges on strengthening these ties. In a three-year Liberian experiment across 197 schools, I evaluate school-based agricultural extension (SBAE) against control, cross-randomizing whether to inform parents about SBAE’s distinctive, inquiry-based learning features via promotional videos and field days. SBAE improves targeted outcomes only when paired with parental engagement: students’ households adopt 26–53% more agricultural techniques, dropouts fall 4–5 percentage points per year, absenteeism falls 28%, and studying hours rise 22%. The program is more cost-effective than most extension programs in Africa and over twice as cost-effective as conditional cash transfers in raising attendance. SBAE has strong positive training effects on students that are unaltered by parental engagement. However, without parental engagement, SBAE’s diffusion mechanisms via school demonstration farms, home projects, and intra-household communication stay muted—technology adoption effects vanish, absenteeism rises 31%, studying hours fall 20%, and parents disconnect—reducing school farm visits 12% and parent–teacher meetings 39%. Parents’ persistent avoidance of information about a new program, plus sharply reversed responses to SBAE under engagement, pose a puzzle. I discuss how standard Bayesian updating under a fixed prior, selective attention, and associative memory each fall short as standalone explanations. I propose an integrative explanation: parental engagement shifts parents’ categorization of SBAE away from schemas such as “school gardens as drudgery.”

WORKING PAPERS

1. “Technology Diffusion and Schooling as Complements: The Promise of School-Based Agricultural Extension in Africa” (joint with Chris Udry)

Efforts to raise agricultural productivity in sub-Saharan Africa are limited by substantial heterogeneity in which technologies are suitable across farms, while low perceived returns to schooling drive students to drop out. We develop a theoretical model of why school-based agricultural extension (SBAE) is suited to address this dual problem. The model extends the target-input framework with two signals of technological suitability, from a school demonstration plot and a home project, both increasing in

precision with the student's study effort. Three results follow. First, schooling and adoption become complements: the same information raises both the value of the new technology and the value of attendance. Second, schooling acquires an option value on the farm: by raising the probability of future adoption, it makes attendance optimal even where the labor-market returns to schooling are low. Third, SBAE addresses suboptimal experimentation by decoupling learning about a technology's suitability from its use in production: because information is acquired through study effort, the returns to schooling pay part of the cost of learning. We provide experimental evidence from Liberia that SBAE cost-effectively raises both agricultural technology adoption and school attendance, and use extensions of the model to study how SBAE's design can be strengthened—including (i) how effectively agriculture and science education are integrated through inquiry-based learning, and (ii) how effectively SBAE's home visits aggregate households' dispersed knowledge and feed it back to them.

2. **“Endogenous Model Uncertainty and Curiosity under Cognitive Limits”**

People actively assign meaning to what they see and predict from it, and whether they look for more information depends on the mental model they use. I model contextual shifts in “mindsets” as a choice of how much model uncertainty to entertain: from a default mental model and a menu of latent, imprecise, and potentially more promising alternatives, an agent chooses how much attention to pay to each and how much information to acquire. This choice is disciplined by a microfounded evidence constraint that bounds motivated beliefs: the weight assigned to an alternative is capped by its likelihood relative to the default and to the other alternatives. Curiosity then arises endogenously as a drive to resolve the model uncertainty that evidence for a more promising alternative creates. The model reproduces regularities of neural remapping and unifies classic belief-updating anomalies with the information-acquisition behavior they usually sit apart from, with two principles for how curiosity can be stimulated. First, contrast matters because an alternative is activated only when the evidence makes the default indefensible against it, and thresholds govern how attention shifts: gradually, abruptly, or not at all. Second, concentration of evidence matters because several alternatives may each beat the default and yet none be activated, however strong the evidence. I apply the framework to the design of information interventions for a school-based agricultural extension program in Liberia, and to pedagogy and learning among teachers, students, and farmers.

3. **“Communicating Parents’ Beliefs to Students as a Pivotal Component of Parental Engagement: Evidence from School-Based Agricultural Extension in Liberia”**

(Funded by NSF, Weiss Fund, Global Poverty Research Lab, and the Buffett Institute for Global Affairs)

When school programs require students to introduce improved techniques on household farms, is informing students and parents about the program sufficient, or is there a case for further signaling parents’ support to students? I study this question within a randomized evaluation of school-based agricultural extension (SBAE) in Liberia, where students take school farm training into a home farming project. After parents view a promotional video of the program and students are encouraged to diffuse techniques to their parents, I randomize a nudge revealing to students their parents’ optimism about their farm-management skills. Two years later, SBAE improves students’ livelihoods only with the nudge: income rises 63%, on-time school fee payment 43–47%, and cost-effectiveness for attendance doubles. Analysis of mechanisms highlights a management aspect of agricultural technology adoption that can be addressed with nudges: even where parents become attentive to SBAE and adopt promoted techniques, third-party assurance is pivotal in shifting students away from assisting in subsistence farming.

WORK IN PROGRESS

Ongoing Randomized Experiments (scaling SBAE across Liberia, Côte d’Ivoire, and Kenya in 2026-35)

1. **“Learning to Learn: Synergizing Inquiry-based Science and Agricultural Transformation in Liberia”** (joint with Vesall Nourani, Harvard GSE; scheduled RCT within the scaling of SBAE in 2026-29)
2. **“Parental Engagement in School-based Agricultural Extension: Testing Mechanisms at Scale”** (piloted interventions in 2025-26)
3. **“Parental Engagement and the Use of Unconditional Cash Transfers”**
4. **“Directed Technical Change under Data-Driven Farming Advice: An Empirical Framework”** (grant from Weiss Fund; piloted survey method; planning RCT)

GRANTS AND AWARDS

Total program evaluation and academic research grants secured for SBAE in Liberia: \$4.2 million

Program Evaluation Grants

AFD Fund for Innovation in Development (\$1.2 mil; co-PI)	2025
J-PAL Learning for All Initiative: Scaling Award (\$222K; co-PI)	2024

USAID Development Innovation Ventures (\$395K; co-PI)	2022
Wellspring Foundation (\$192K; co-PI)	2022
AFD Fund for Innovation in Development (\$1.2 mil; co-PI)	2021
Food and Agriculture Organization (\$68K; co-PI)	2021

Academic Research Awards

J-PAL Learning for All Initiative (\$400K; co-PI)	2025
Weiss Fund (\$5K; Lead PI)	2022
National Science Foundation (\$451K; co-PI)	2020
Weiss Fund (\$29K; Lead PI)	2020
Global Poverty Research Lab (\$19K; Lead PI)	2019
Buffett Institute for Global Affairs (\$5K; Lead PI)	2019

INVITED PRESENTATIONS

During postdoctoral studies:

National Chung Cheng University; NYU Africa House—Center for Technology, Economics, and Development (CTED); Mohammed VI Polytechnic University (UM6P); Tel Aviv University; Oklahoma State University; Harvard Center for International Development; Africa Food Systems Forum 2024

During PhD:

Meeting of the Society of Economics of the Household (SEHO); J-PAL—Middle East and North Africa (MENA); CGIAR—Standing Panel on Impact Assessment (SPIA); International Food Policy Research Institute (IFPRI); University of Glasgow; National Graduate Institute for Policy Studies (GRIPS, Tokyo); Development Day 2023

TEACHING

Teaching Assistant, Northwestern University (Department median 5.3/6)

Introduction to Microeconomics (5.4/6), Introduction to Macroeconomics (5.4/6), Intermediate Microeconomics (5.4/6), Intermediate Macroeconomics (5.7/6), International Trade (5.6/6), Economics of Nonprofit Organizations (5.8/6)

Teaching Assistant, The Chinese University of Hong Kong

Mathematical Methods in Economics I, II, Family Economics, International Macroeconomics

LANGUAGES AND SKILLS

Citizenship: Hong Kong Special Administrative Region, China

Languages: English (fluent), French (intermediate), Mandarin (fluent), Cantonese (native)

Programming: Stata, R, Matlab, Python, ArcGIS, SurveyCTO

REFERENCES

Professor Christopher Udry (Chair)

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Professor Lori Beaman

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